WASP-11b

R-0753

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_231205` · created 2026-07-28T03:11:58+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460283.8478 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.047
Transit depth (Rp/Rs)^2	2.54 +/- 1.48 [%]
Orbital Inclination (inc)	85.18 +/- 1.69
Airmass coefficient 1 (a1)	1.132 +/- 0.044
Airmass coefficient 2 (a2)	-0.09 +/- 0.019
Scatter in the residuals of the lightcurve fit is	3.83 %
Best Comparison Star	#3 - [153, 152]
Optimal Aperture	3.95
Optimal Annulus	8.9
Transit Duration (day)	0.046 +/- 0.039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.047 vs 0.131 (deviation 0.62σ)
Duration fit vs published	0.046 d vs 0.1075 d (deviation 1.58σ)
Mid-time O–C	1.9 min
Depth SNR	3.4
Residual scatter	3.83 %

Light curve

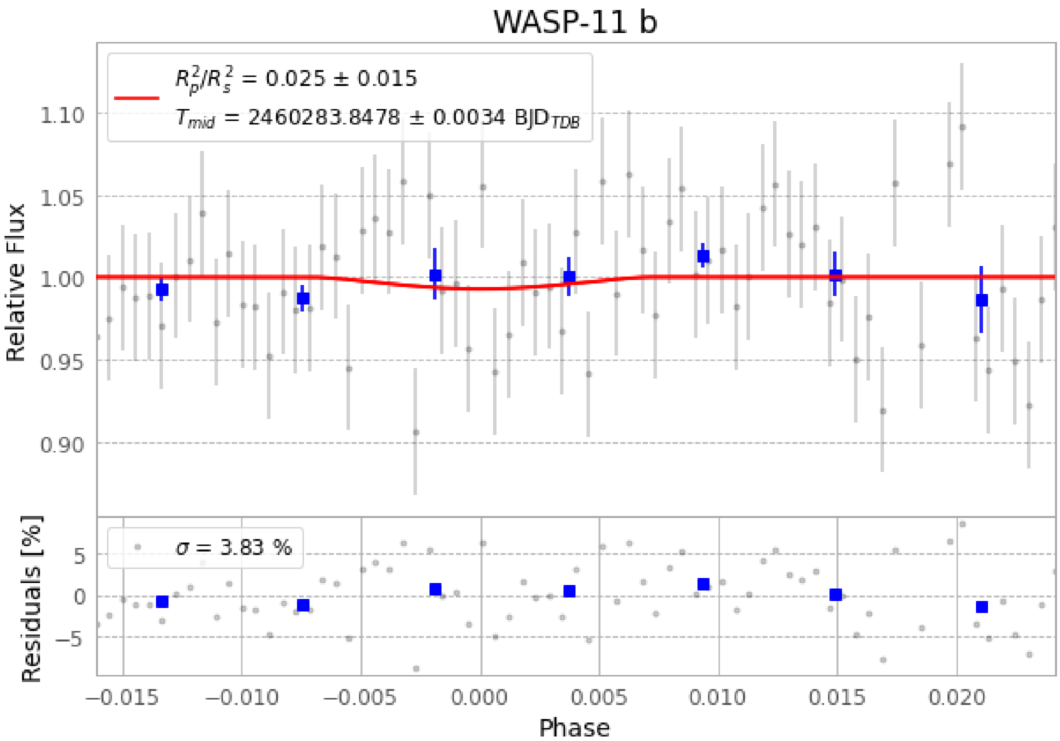

Figure 1 — FinalLightCurve_WASP-11 b_2023-12-04.png (SHA-256 7d586a96d078e008…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [350, 333], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 7f873fd73ea18868…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

75 FITS frames (49 MB), `HATP-10231205064516.FITS` → `HATP-10231205102715.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-231205.log (055b5f930d3b…), FinalLightCurve_WASP-11 b_2023-12-04.png (7d586a96d078…), FinalLightCurve_WASP-11 b_2023-12-04.csv (a58b957a8987…)

Compute resources

Wall time 375.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)